DAY 41

Monitoring Machine Learning Models in Production

Latency, errors, data drift, concept drift, and prediction quality

Learning outcome: Learn what to monitor after deployment and practice recording prediction results, timing, errors, and simple quality/drift indicators.

1. Why monitoring matters

Deploying an ML model is not the end. Real inputs, software dependencies, and user behavior can change. Monitoring helps identify slow responses, failures, unusual inputs, and possible changes in prediction quality.

2. Four monitoring areas

API performance: response latency, availability, request volume, and error rate.
Input data: missing values, invalid values, and changing feature distributions.
Model quality: metrics from comparing predictions with actual outcomes when labels become available.
Operational errors: exceptions, timeouts, failed requests, and dependency problems.

3. Data drift vs concept drift

Data drift: production input distribution differs from a reference distribution.
Concept drift: the relationship between inputs and the target changes.
Performance degradation: measured model quality becomes worse on relevant evaluation data.

Data drift does not automatically prove model quality has dropped. It is a signal to investigate. Actual quality checks usually require labeled outcomes.

4. Measure prediction latency and errors

time.perf_counter() measures elapsed time. The practice file records response_time_ms and success/error status. It records the exception type rather than exposing raw internal error text.

```
start = time.perf_counter()
try:
    prediction = model.predict([[study_hours, attendance]])
    elapsed_ms = (time.perf_counter() - start) * 1000
    record = {"status": "success", "response_time_ms": elapsed_ms}
except Exception as error:
    elapsed_ms = (time.perf_counter() - start) * 1000
    record = {"status": "error",
              "error_type": type(error).__name__,
              "response_time_ms": elapsed_ms}
```

5. Mean Absolute Error (MAE)

MAE is the average absolute difference between actual and predicted values. Lower MAE means smaller average absolute errors on the evaluated sample.

```
MAE = sum(abs(actual - predicted)) / number of predictions
```

Example: actual [70, 80, 90], predicted [65, 85, 90]. Absolute errors are [5, 5, 0]. MAE = 10/3 = 3.33.

6. Simple mean-shift indicator

The Python file calculates the absolute difference between the training-sample mean and production-sample mean. A large difference can flag a possible shift. This is only a basic educational indicator, not a formal statistical drift test, and it cannot prove performance degradation.

```
shift = abs(mean(training_values) - mean(production_values))
```

7. Run the practice file

```
python -m pip install pytest\npytest -v Day-41-ML-Model-Monitoring-Practice.py
```

The example uses a DemoModel, so no saved model file is needed. Tests cover successful prediction monitoring, error handling, MAE, and mean-shift calculation.

8. Production best practices

• Avoid logging passwords, tokens, or unnecessary personal data.
• Do not expose stack traces or internal exception messages to API users.
• Compare distributions over meaningful time windows.
• Choose alert thresholds based on service needs.
• Investigate drift signals and confirm quality with labels where possible.
• Keep monitoring records secure and structured.

9. Practice tasks with solutions

Task 1: Actual [70,80,90], predicted [65,85,90]. **Solution:** Errors [5,5,0], MAE = 3.33.
Task 2: Why measure latency? **Solution:** To identify slow requests and track service expectations.
Task 3: Does drift prove accuracy fell? **Solution:** No, it is a signal to investigate.
Task 4: Why not return raw exception details? **Solution:** They may reveal internal implementation details.

10. Day 41 checklist

■ I can explain latency and error monitoring.
■ I understand data drift and concept drift.
■ I can calculate MAE.
■ I know drift is not automatic proof of poor model quality.
■ I understand why logs must protect sensitive data.

GitHub learning repository: https://github.com/DeveloperDesigner-star/Datasciencepdfs